

Extra Practice — Answer Key

Pandas (group_by, pivot, indexing) & OOP. Every answer verified by running the code.

Quick grid

Q1: b

Q2: c

Q3: a

Q4: d

Q5: b

Q6: c

Q7: a

Q8: d

Q9: b

Q10: c

Q11: a

Q12: d

Q13: b

Q14: c

Q15: a

Detailed answers

Q1. Answer: b

Output:

```
Dog named Rex  
[Dog(Rex)]
```

Why: print() uses __str__; showing an object inside a list uses each element's __repr__.

Q2. Answer: c

Output:

```
2
```

Why: Each bump() adds 1 to this instance's count attribute; two calls make it 2.

Q3. Answer: a

Output:

```
$8
```

Why: __add__ returns a new Money(8); print() then uses __str__, which formats it as \$8.

Q4. Answer: d

Output:

```
meow
```

```
...
```

Why: Cat overrides speak(); Fish defines nothing, so it inherits Animal's speak() -> '...'.

Q5. Answer: b

Output:

```
3 / 20
```

Why: __len__ makes len(b) return 3; __getitem__ makes b[1] return the item at index 1 (20).

Q6. Answer: c

Output:

```
99
```

Why: Objects are mutable and passed by reference, so modify() changes the real instance -> 99.

Q7. Answer: a

Output:

```
team
```

```
A    40
```

```
B    60
```

```
Name: pts, dtype: int64
```

Why: groupby sums pts within each team: A = 10+30 = 40, B = 20+40 = 60.

Q8. Answer: d

Output:

```
g
```

```
x    3
```

```
y      1
dtype: int64
```

Why: size() counts the rows in each group: x appears 3 times, y once.

Q9. Answer: b

```
Output:
6.0
```

Why: Group 'a' mean = $(4+8)/2 = 6.0$; .loc['a'] pulls that value.

Q10. Answer: c

```
Output:
dept
eng      110
sales     70
Name: pay, dtype: int64
```

Why: max pay per dept; groupby sorts the group keys alphabetically, so eng (110) comes before sales (70).

Q11. Answer: a

```
Output:
sales
day
Mon      30
Tue      40
```

Why: pivot_table with aggfunc='sum' totals sales per day: Mon = 10+20 = 30, Tue = 40.

Q12. Answer: d

```
Output:
col  c1  c2
row
r1   1.0  2.0
r2   3.0 NaN
```

Why: pivot reshapes to a row x col grid; the missing (r2, c2) cell becomes NaN, which upcasts the whole thing to float.

Q13. Answer: b

```
Output:
n
cat
x      4.0
y     10.0
```

Why: pivot_table's DEFAULT aggfunc is mean, so $x = (2+6)/2 = 4.0$, $y = 10.0$.

Q14. Answer: c

```
Output:
b    p    q
a
x   1.0  2.0
y   3.0 NaN
```

Why: Grouping by two keys makes a MultiIndex; unstack moves 'b' to columns, and the missing (y, q) cell becomes NaN (float).

Q15. Answer: a

```
Output:
c      3.0
a      1.0
d      NaN
dtype: float64
```

Why: reindex reorders to the new label order and inserts NaN for the new label 'd', upcasting to float.